



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA0905 JAVA PROGRAMMING – C02-AT1

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1.

```
Your Code
1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         String paragraph = sc.nextLine();
6         int characters = paragraph.length();
7         int words = paragraph.trim().isEmpty() ? 0 : paragraph.trim().split("\\s+").length;
8         int sentences = paragraph.split("[.!?]*").length;
9         String oldWord = sc.nextLine();
10        String newWord = sc.nextLine();
11        String replaced = paragraph.replace(oldWord, newWord);
12        String[] s = replaced.split("(?<[.!?])");
13        StringBuilder finalDoc = new StringBuilder();
14        StringBuffer history = new StringBuffer();
15        history.append("Original: ").append(paragraph).append("\n");
16        history.append("Replaced: ").append(oldWord).append(" with ").append(newWord).append("\n");
17        for (String str : s) {
18            String t = str.trim();
19            String rev = new StringBuilder(t).reverse().toString();
20            finalDoc.append(rev).append(" ");
21        }
22
23        System.out.println("Original Document:");
24        System.out.println(paragraph);
25        System.out.println("Characters: " + characters);
26        System.out.println("Words: " + words);
27        System.out.println("Sentences: " + sentences);
28        System.out.println("Editing History:");
29        System.out.println(history);
30        System.out.println("Final Document:");
31        System.out.println(finalDoc.toString().trim());
32    }
33 }
```

Input

Artificial Intelligence is the future.
AI is changing the world.
AI
Machine Learning

Output

is changing the world.
Replaced 'AI' with 'Machine Learning'
Final Document:
dnuof eht si eehgillelnf

2.

```
Your Code
1 import java.util.*;
2 public class Main {
3     static class Patient {
4         int id;
5         String name;
6         String department;
7         Patient(int id, String name, String department) {
8             this.id = id;
9             this.name = name;
10            this.department = department;
11        }
12        public String toString() {
13            return id + " " + name + " " + department;
14        }
15    }
16    public static void main(String[] args) {
17        Scanner sc = new Scanner(System.in);
18        LinkedList<Patient> list = new LinkedList<>();
19        int n = sc.nextInt();
20        for (int i = 0; i < n; i++) {
21            int id = sc.nextInt();
22            String name = sc.next();
23            String dept = sc.next();
24            list.add(new Patient(id, name, dept));
25        }
26        System.out.println("Patients in Registration Order:");
27        for (Patient p : list)
28            System.out.println(p);
29        int updateId = sc.nextInt();
30        String newName = sc.next();
31        String newDept = sc.next();
32        for (Patient p : list) {
33            if (p.id == updateId) {

```

Input

4
101 Rahul Cardiology
102 Priya Neurology
103 Arun Orthopedics

Output

Patients in Registration Order:
101 Rahul Cardiology
102 Priya Neurology
103 Arun Orthopedics
104 Sneha Cardiology
Enter Patient's Id to update:

3.

Your Code

```

1 import java.util.*;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         Queue<String> queue = new LinkedList<>();
6         PriorityQueue<String> vipQueue = new PriorityQueue<>();
7         int n = sc.nextInt();
8         sc.nextLine();
9         for (int i = 0; i < n; i++) {
10             queue.add(sc.nextLine());
11         }
12         System.out.println("Current Queue:");
13         System.out.println(queue);
14         System.out.println("Next Passenger:");
15         System.out.println(queue.peek());
16         System.out.println("Boarding Passengers:");
17         while (!queue.isEmpty()) {
18             System.out.println("Boarded: " + queue.poll());
19             System.out.println("Remaining Queue: " + queue);
20         }
21         int m = sc.nextInt();
22         sc.nextLine();
23         for (int i = 0; i < m; i++) {
24             vipQueue.add(sc.nextLine());
25         }
26         System.out.println("VIP Boardings:");
27         while (!vipQueue.isEmpty()) {
28             System.out.println(vipQueue.poll());
29         }
30     }
31 }

```

Input

```

4
Rahul
Priya
Arun

```

Output

```

Current Queue:
[Rahul, Priya, Arun, Sneha]
Next Passenger:
Rahul
Boarding Passengers:
Boarded: Rahul

```

4.

Your Code

```

1 import java.util.LinkedHashSet;
2 import java.util.TreeSet;
3 public class Main {
4     public static void main(String[] args) {
5         LinkedHashSet<String> certificates = new LinkedHashSet<String>();
6         certificates.add("CERT101");
7         certificates.add("CERT102");
8         certificates.add("CERT103");
9         certificates.add("CERT101");
10        System.out.println("Insertion Order:");
11        System.out.println(certificates);
12        if (certificates.contains("CERT102")) {
13            System.out.println("CERT102 Verified");
14        } else {
15            System.out.println("Not Found");
16        }
17        TreeSet<String> sorted = new TreeSet<String>(certificates);
18        System.out.println("Sorted Order:");
19        System.out.println(sorted);
20        System.out.println("Total Certificates: " + certificates.size());
21        certificates.remove("CERT103");
22        System.out.println("After Removal:");
23        System.out.println(certificates);
24    }
25 }

```

Input

```

stdin input

```

Output

```

Insertion Order:
[CERT101, CERT102, CERT103]
CERT102 Verified
Sorted Order:
[CERT101, CERT102, CERT103]
Total Certificates: 3

```

Run

Save

5.

Your Code

```

1 import java.util.HashMap;
2 import java.util.Map;
3 public class Main {
4     public static void main(String[] args) {
5         HashMap<Integer, String> product = new HashMap<>();
6         HashMap<Integer, Integer> quantity = new HashMap<>();
7         HashMap<Integer, Integer> price = new HashMap<>();
8         HashMap<Integer, String> category = new HashMap<>();
9         product.put(101, "Laptop");
10        quantity.put(101, 10);
11        price.put(101, 50000);
12        category.put(101, "Electronics");
13        product.put(102, "Mouse");
14        quantity.put(102, 20);
15        price.put(102, 500);
16        category.put(102, "Accessories");
17        quantity.put(101, 15);
18        int id = 101;
19        if (product.containsKey(id)) {
20            System.out.println("Product Found");
21            System.out.println("ID: " + id);
22            System.out.println("Name: " + product.get(id));
23            System.out.println("Quantity: " + quantity.get(id));
24        }
25        int total = 0;
26        for (Integer key : product.keySet()) {
27            total += quantity.get(key) * price.get(key);
28        }
29        System.out.println("Total Inventory Value: " + total);
30        System.out.println("\nProducts Category-wise:");
31        for (Integer key : product.keySet()) {

```

Input

```

stdin input

```

Output

```

Product Found
ID: 101
Name: Laptop
Quantity: 15
Total Inventory Value: 760000

```